

Who Wins and Who Loses In Prediction Markets?

Evidence from Polymarket

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University of Alberta · June 3, 2026

Prediction markets have exploded in popularity

- Platforms like Polymarket and Kalshi let users trade on the outcomes of real-world events: politics, economics, sports, entertainment.
- Now routinely cited in major news outlets alongside polls and analyst forecasts; media partnerships are proliferating.
- Fintech and blockchain let them scale fast: \$67B traded on Polymarket and \$53B on Kalshi over the last four years.
- Their growth has prompted concerns about the gamification of trading, prediction markets as an entry point for young/retail investors, and insider trading.

Trading is zero-sum: who bears the losses?

- Contracts are binary contingent claims that pay \$1 / \$0 on resolution. Trading is zero-sum: every dollar gained is a dollar lost.
- Unlike equities: no equity premium, no dividends, no passive option.
- Short maturities encourage short-horizon trading, precisely what behavioral finance flags as most destructive to wealth (Odean, 1999; Barber and Odean, 2000).

Research question

Who wins and who loses in prediction markets?

Five main findings

1. **Profits are concentrated**: top 0.1% capture 51.2% of gains, top 1% capture 76.5%; **69.0% of users lose money**.
2. **Prices are well calibrated** on aggregate (a contract at p resolves yes $\approx p\%$ of the time), though thin/low-attention markets deviate.
3. **Excess hit rate** falls sharply down the PnL distribution (+20 pp at the top to -15 to -18 pp in the loss tail).
4. **Liquidity provision** is the strongest cross-sectional predictor of avoiding losses (a 1-SD higher maker share: -9.3 pp prob. of loss).
5. The 100 most successful users (27.7% of winners' profits) look **unlikely to be insiders**: gains span many sports and a few election markets, with a large share from pure liquidity provision.

How Polymarket Works



Sports · Hockey

2026 NHL Stanley Cup Champion



\$81,025,582 Vol. | Jun 30, 2026

1H 6H 1D 1W 1M ALL

Carolina Hurricanes

\$1,310,892 Vol.

59% ▲ 38%

Buy Yes 59¢

Buy No 42¢

Vegas Golden Knights

\$2,601,048 Vol.

41% ▲ 21%

Buy Yes 41.2¢

Buy No 58.9¢

View resolved

Rules Market Context

This market will resolve to "Yes" if the Carolina Hurricanes win the 2026 NHL Stanley Cup. Otherwise, this market will resolve to "No".

This market will resolve to "No" if it becomes impossible for this team to win the 2026 NHL Stanley Cup based off the rules of the NHL.

The resolution source for this market will be information from the NHL.

Comments (115) [Top Holders](#) [Positions](#) [Activity](#)

Add a comment...



Post



2026 NHL Stanley Cup Champion
Carolina Hurricanes · Yes

Buy Sell

Limit

Yes 59¢

No 42¢

Limit price

- 59¢ +

Shares

0

-100

-10

+10

+100

Expires

24h

Total

\$0.00

To win

📈 \$0.00

Trade

By trading, you agree to the [Terms of Use](#).

All

Sports

NHL



Will the San Antonio Spurs win the 2026 NBA Finals?

64%



Will Frederik Andersen win the 2026 NHL Conn Smythe Trophy?

35%



Will Andrei Vasilevskiy win the 2025-2026 NHL Vezina Trophy?

92%



2026 NHL Stanley Cup Champion



🔄 \$81,025,582 Vol. | 🕒 Jun 30, 2026

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Buy No 42¢

Order Book Graph Resolution

💰 Maker Rebate + Rewards 🔄

TRADE YES 📄

PRICE

SHARES

TOTAL

	62¢	2,650.00	\$57,622.62
	61¢	40,303.00	\$55,979.62
	60¢	2,904.72	\$31,394.79
Asks	59¢	50,257.56	\$29,651.96

Last: 59¢

Spread: 1¢

Bids	58¢	95,303.66	\$55,276.12
	57¢	11,000.00	\$61,546.12
	56¢	3,083.61	\$63,272.94
	55¢	44,084.68	\$87,519.51

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2026 NHL Stanley Cup Champion

Carolina Hurricanes · Yes

Buy Sell

Market ▾

Yes 59¢

No 42¢

Amount

\$0

+\$1

+\$5

+\$10

+\$100

Trade

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Sports

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2026 NHL Stanley Cup Champion



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Carolina Hurricanes

\$1,310,892 Vol.

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Buy Yes 59¢

Buy No 42¢

Order Book Graph Resolution

🏆 Maker Rebate + Rewards 🔄

TRADE NO 📄

PRICE

SHARES

TOTAL



Last: 41¢

Spread: 1¢

Bids

41¢	50,257.56	\$20,605.60
40¢	2,904.72	\$21,767.49
39¢	40,303.00	\$37,485.66
38¢	2,650.00	\$38,492.66

Vegas Golden Knights

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41% ▲ 21%

Buy Yes 41.2¢

Buy No 58.9¢

View resolved ▾

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2026 NHL Stanley Cup Champion

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All

Sports

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Trading: a hybrid CLOB with on-chain settlement

- Decentralized platform; trades execute and settle via smart contracts on the Polygon blockchain (no central intermediary).
- Hybrid architecture: a central limit order book matches trades, conditional tokens settle on-chain.
- Users post limit orders (makers) or trade against existing liquidity (takers). Each user is a wallet address; every trade is permanently, publicly recorded.
- Tick size \$0.01, finer (\$0.001) below \$0.04 or above \$0.96.
- No trading fees on most markets in our sample, so makers earn the spread and takers pay it.

Settlement and resolution

- Each event is a set of binary markets; e.g. “2026 Stanley Cup Champion” has one **Yes/No** market per team.
- A **Yes** token pays \$1 if the outcome occurs, \$0 otherwise; a **No** token pays the complement. Denominated in USDC.
- Settlement is automatic: on resolution the smart contract assigns \$1 to winners and \$0 to losers. No counterparty risk.
- Resolution via the UMA oracle: an outcome is proposed, then a challenge window during which anyone can dispute by posting a bond; UMA token holders vote, the losing side forfeits the bond.

Three caveats to keep in mind

1. **Prices = probabilities only under risk-neutral pricing.** Ignoring risk preferences and time value, a price of \$0.35 means a 35% implied probability, an assumption rarely tested (Manski, 2006).
2. **We observe only Polymarket activity.** We cannot detect arbitrage or other multi-platform strategies run off-platform, which may matter for sophisticated traders.
3. **We observe “wallets”, not individuals.** One individual can control many wallets.

Literature

Literature: information aggregation and the marginal trader

- Prices coordinate dispersed knowledge (Hayek, 1945); real-money markets beat polls (Forsythe et al., 1992); canonical survey (Wolfers and Zitzewitz, 2004).
- Accuracy does not require individual rationality: a small group of informed marginal traders setting limit orders drives prices to efficiency (Forsythe et al., 1992).
- Price-takers violate no-arbitrage far more than makers (Oliven and Rietz, 2004).
- Closest to us: Barber et al. (2009), a few sophisticated traders profit at retail's expense; similar heterogeneity in options (Hu et al., 2024) and futures (Dewally et al., 2013).

Literature: behavioral biases and recent Polymarket work

- Favorite-longshot bias (Snowberg et al., 2013); excessive trading loses money (Barber and Odean, 2000). Polymarket users disproportionately trade extreme prices.
- Prediction markets have spurred a recent wave: 2024 election microstructure (Tsang and Yang, 2026); cross-platform price discovery (Ng et al., 2026); maker-taker wealth transfer on Kalshi (Becker, 2025); earnings expectations from prices (Gomez Cram et al., 2025); wash-trading detection (Sirolly et al., 2025).
- Our contribution: the first wallet-level accounting of the full distribution of profits and losses, and the role of maker-taker dynamics. We caution on skill claims given selection in who keeps trading (cf. Gomez Cram et al., 2026).

Data

Data: sources and trade reconciliation

We assemble the complete on-chain transaction history from two sources:

- Metadata (events, markets, tags, winning outcome): Polymarket's Gamma and CLOB APIs.
- Trades (on-chain order fills, with maker/taker wallets): the Goldsky SubGraph (GraphQL over Polygon).

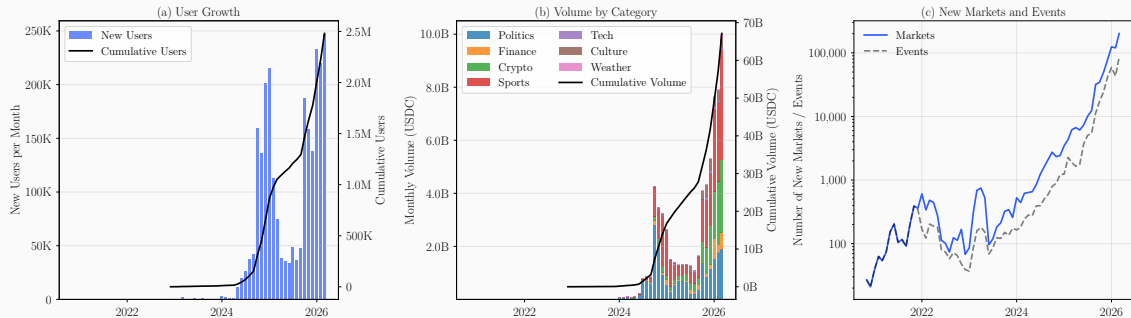
We reconcile the two into one record per trade (price, quantity, direction, counterparties, resolved outcome), the full population of trades. Each trade is normalized to a reference outcome, so price p is the implied probability.

The complete history spans [Nov 11, 2022 – Mar 29, 2026](#): [2.4M users](#), [588M trades](#), [\\$67B volume](#) (ending just before platform-wide fees on Mar 30, 2026).

Categories. Polymarket has no clean category field. We assign each market to one of seven categories (Sports, Crypto, Politics, Finance, Tech, Culture, Weather) through a hierarchical cascade: event tags, then a legacy category string, title heuristics, and manual overrides. Every market lands in exactly one category; none is dropped.

Resolution. Markets settle on-chain via the UMA oracle: the winning token pays \$1, losers \$0. We take the winning outcome from the Gamma API (with a settlement-price fallback) and mark positions daily, so $PnL = \text{portfolio value} + \text{net USDC balance}$.

Platform growth: users, volume, new markets



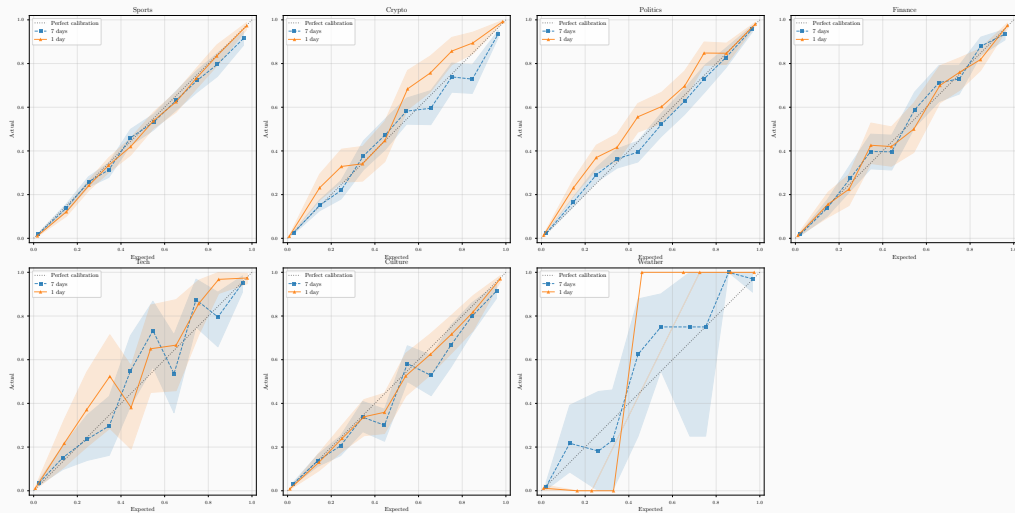
Trading activity is highly skewed across users

Metric	Mean	Std. Dev.	Min	5th Pct.	25th Pct.	Median	75th Pct.	95th Pct.	Max
Panel A: All users (N = 2, 480, 104)									
Trades per User	474	21,450	1	1	9	28	74	546	13,587,640
Markets per User	54	495	1	1	3	10	26	127	130,387
Volume per User (USDC)	54,143	1,699,528	0	20	474	2,343	11,619	97,793	604,671,119
Fraction Extreme Price	54.9%	39.7%	0.0%	0.0%	14.0%	58.3%	100.0%	100.0%	100.0%
PnL per User (USDC)	0	32,373	-10,022,403	-920	-32	-2	1	251	22,029,973
Fraction Maker Volume	17.9%	27.2%	0.0%	0.0%	0.0%	0.0%	39.5%	75.4%	100.0%
Panel B: Users with more than 100 trades (N = 478, 096)									
Trades per User	2,349	48,809	101	107	137	221	576	4,552	13,587,640
Markets per User	231	1,110	1	9	31	61	132	800	130,387
Volume per User (USDC)	234,116	3,863,332	13	1,129	4,557	15,411	50,156	447,591	604,671,119
Fraction Extreme Price	48.7%	35.2%	0.0%	1.7%	17.1%	39.7%	87.3%	100.0%	100.0%
PnL per User (USDC)	181	73,603	-10,022,403	-2,688	-147	-14	5	1,183	22,029,973
Fraction Maker Volume	24.2%	27.9%	0.0%	0.0%	0.1%	9.3%	47.3%	81.0%	100.0%
Panel C: Users with more than 1,000 trades (N = 78, 366)									
Trades per User	12,957	119,998	1,001	1,070	1,473	2,509	5,673	29,905	13,587,640
Markets per User	1,023	2,594	1	45	189	433	971	3,583	130,387
Volume per User (USDC)	1,150,361	9,464,494	152	6,920	24,393	78,459	329,973	3,111,043	604,671,119
Fraction Extreme Price	31.5%	29.1%	0.0%	1.3%	9.5%	20.6%	45.7%	96.7%	100.0%
PnL per User (USDC)	2,632	179,101	-10,022,403	-11,852	-916	-141	122	12,631	22,029,973
Fraction Maker Volume	31.9%	31.2%	0.0%	0.0%	1.6%	23.5%	55.4%	91.9%	100.0%

Activity by market category

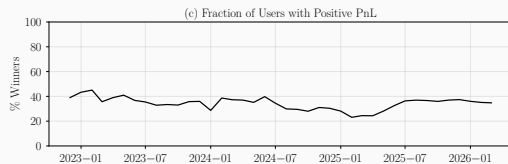
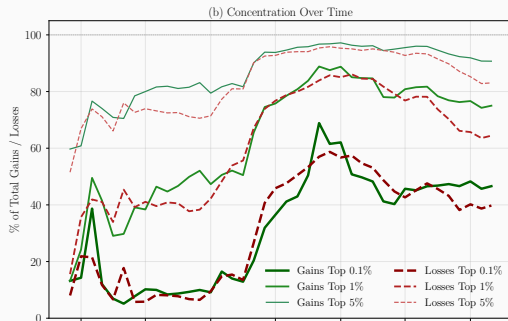
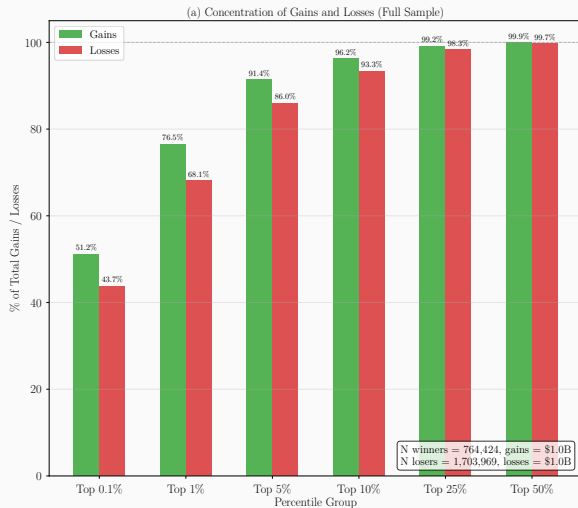
Category	Events	Markets	Trades (M)	Trades (%)	Volume (M USDC)	Volume (%)	Users	Users (%)	Trades/Markets	Volume/Markets
Sports	64,205	321,231	95.4	16.2	25,477.0	37.9	1,573,302	63.4	296.9	79,311
Politics	7,248	29,926	55.3	9.4	21,114.4	31.5	1,568,028	63.2	1,847.6	705,553
Crypto	172,313	206,491	394.7	67.1	12,762.0	19.0	1,373,525	55.4	1,911.4	61,804
Culture	2,530	15,806	19.5	3.3	3,473.9	5.2	732,477	29.5	1,235.0	219,781
Finance	5,539	17,159	9.2	1.6	3,107.5	4.6	720,184	29.0	534.7	181,100
Tech	1,016	4,308	5.9	1.0	806.9	1.2	469,579	18.9	1,363.6	187,303
Weather	2,576	19,962	8.3	1.4	393.7	0.6	199,565	8.0	417.7	19,725
Total	255,427	614,883	588.3	100.0	67,135.4	100.0	2,480,080	100.0	956.7	109,184

Prices are well calibrated on aggregate



The Cross-Section of Profits

Gains and losses are highly concentrated



PnL concentration by category

	All	Sports	Politics	Crypto	Culture	Finance	Tech	Weather
Descriptive Statistics								
N Markets	614,883	321,231	29,926	206,491	15,806	17,159	4,308	19,962
N Events	255,427	64,205	7,248	172,313	2,530	5,539	1,016	2,576
N Users (000s)	2,480.1	1,573.3	1,568.0	1,373.5	732.5	720.2	469.6	199.6
N Trades (M)	588.3	95.4	55.3	394.7	19.5	9.2	5.9	8.3
Volume (M USDC)	67,135.4	25,477.0	21,114.4	12,762.0	3,473.9	3,107.5	806.9	393.7
Frac. Winners	30.8%	31.3%	35.3%	42.4%	38.4%	46.4%	50.1%	52.5%
Mean PnL by Percentile – Losers (PnL < 0)								
0-20	-0.38	-0.22	-0.23	-1.09	-0.08	-0.11	-0.09	-0.42
20-40	-3.25	-1.91	-2.13	-10	-0.88	-1.10	-1.11	-3.59
40-60	-14	-11	-9.41	-41	-5.35	-6.73	-6.70	-14
60-80	-61	-56	-53	-184	-30	-33	-31	-48
80-90	-257	-252	-277	-759	-133	-148	-119	-154
90-95	-893	-866	-1,035	-2,033	-483	-534	-388	-430
95-99	-2,763	-3,142	-4,905	-6,184	-2,363	-2,481	-1,673	-1,624
99-99.9	-16,682	-20,193	-37,200	-27,875	-19,286	-18,946	-10,711	-8,168
99.9-100	-269,192	-328,420	-442,567	-176,141	-264,698	-199,901	-99,020	-53,277
Mean PnL by Percentile – Winners (PnL > 0)								
0-20	0.19	0.09	0.08	0.10	0.03	0.03	0.02	0.01
20-40	2.17	2.40	0.91	1.17	0.44	0.25	0.25	0.17
40-60	13	13	5.04	7.12	3.05	1.93	1.55	1.30
60-80	73	62	27	34	15	12	9.18	9.51
80-90	346	297	122	166	55	51	34	34
90-95	1,308	1,116	449	686	164	164	98	98
95-99	5,111	4,067	2,469	4,145	889	887	502	583
99-99.9	38,634	30,777	23,892	23,926	8,479	7,542	5,044	4,043
99.9-100	703,060	597,533	632,805	217,854	165,827	172,044	57,066	50,548

Measuring forecasting skill: the excess hit rate (EHR)

Recast every trade to the buy side: $o_k \in \{0, 1\}$ equals 1 if the contract *bought* in trade k resolves in-the-money, p_k is the buy-side price, and q_k is the traded quantity (shares).

Per-trade EHR

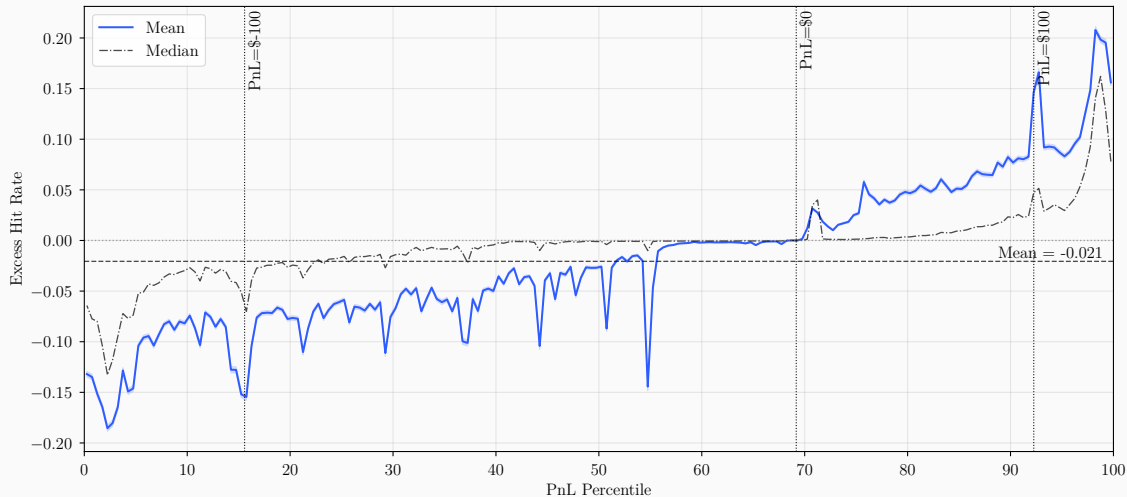
$$o_k - p_k$$

Volume-weighted EHR, user i

$$\text{EHR}_{\text{vw},i} = \frac{\sum_k q_k (o_k - p_k)}{\sum_k q_k}$$

Positive EHR: trades resolve **more favorably** than their prices implied; negative EHR: the opposite.

Forecasting accuracy declines down the PnL distribution



Who Wins and Who Loses?

Probit regression: Pr(Loss)

	All Users				Users With More Than 100 Trades		Users With More Than 1,000 Trades	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Frac Extreme Price	−0.013*** (0.001)	−0.018*** (0.001)	−0.008*** (0.001)	−0.004*** (0.001)	0.004* (0.002)	0.009*** (0.002)	−0.128*** (0.006)	−0.126*** (0.006)
Frac Maker Volume		−0.357*** (0.001)	−0.353*** (0.001)	−0.347*** (0.001)	−0.332*** (0.002)	−0.329*** (0.002)	−0.148*** (0.006)	−0.143*** (0.006)
Log N Trades		−0.000* (0.000)	0.004*** (0.000)	0.017*** (0.000)	0.027*** (0.001)	0.030*** (0.001)	−0.004* (0.002)	−0.006*** (0.002)
Log Total Volume		0.020*** (0.000)	0.018*** (0.000)	0.019*** (0.000)	−0.020*** (0.000)	−0.021*** (0.001)	−0.044*** (0.001)	−0.041*** (0.001)
Category HHI			0.080*** (0.001)	−0.054*** (0.002)	−0.009*** (0.003)	−0.064*** (0.004)	−0.130*** (0.007)	−0.142*** (0.010)
Counterparty HHI			−0.005*** (0.002)	0.035*** (0.002)	0.181*** (0.007)	0.207*** (0.007)	0.506*** (0.050)	0.500*** (0.049)
Category Dummies	No	No	No	Yes	No	Yes	No	Yes
Observations	2,480,104	2,480,104	2,480,104	2,480,104	478,096	478,096	78,366	78,366
Pseudo R ²	0.000	0.035	0.036	0.044	0.042	0.043	0.054	0.056

Main results are also robust to a resolved-markets-only subsample and to fee-free markets.

Risk-adjusting performance: reward-to-risk on monthly PnL

Computed at the user level from each user's monthly mark-to-market PnL series over the full sample. Analogs to Sharpe and Sortino on dollar PnL (returns have no natural capital base on Polymarket).

Annualized PnL-to-volatility

$$\sqrt{12} \frac{\overline{\Delta \text{PnL}}}{\sigma(\Delta \text{PnL})}$$

PnL-to-downside-volatility

$$\sqrt{12} \frac{\overline{\Delta \text{PnL}}}{\text{RMS}(\Delta \text{PnL}^-)}$$

ΔPnL are monthly PnL changes; ΔPnL^- keeps only negative months. Reported as group medians (heavy upper tail).

Characteristics across PnL groups

	PnL Group Mean				Difference		
	Top 0.1% (1)	Next 0.9% (2)	Next 4% (3)	Bot. 95% (4)	0.1%–95% (5)	0.9%–95% (6)	4%–95% (7)
<i>Panel A: Behavioral Characteristics</i>							
Frac Maker Volume	0.4731	0.4145	0.3237	0.1705	0.3026***	0.2439***	0.1532***
Frac Extreme Price	0.2823	0.2864	0.3114	0.5642	-0.2818***	-0.2778***	-0.2528***
Category HHI	0.8032	0.8166	0.7949	0.7670	0.0363***	0.0496***	0.0279***
Avg N Trades	150,623	8,389	1,165	223	150,401***	8,166***	943***
<i>Panel B: Risk-Adjusted Performance</i>							
Mean Monthly PnL (USDC)	54,586.11	2,468.02	310.52	-104.53	54,690.64***	2,572.55***	415.04***
SD Monthly PnL (USDC)	104,020.18	5,097.63	638.99	220.13	103,800.05***	4,877.50***	418.86***
Max Drawdown (USDC)	35,215.77	1,898.77	228.37	365.99	34,849.78***	1,532.78***	-137.62***
Median Ann. PnL-to-Volatility	1.732	1.468	1.459	-1.150	2.882***	2.618***	2.609***
Median Ann. PnL-to-Downside-Volatility	12.079	20.407	12.135	-1.361	13.440***	21.768***	13.496***
Excess Hit Rate	0.0977	0.1319	0.1028	-0.0265	0.1241***	0.1584***	0.1293***

Forecasting Skill and Liquidity Provision

Monthly performance is only weakly persistent

Panel A: 1-Month Horizon

Starting Group	N	< -100k	[-100k, -10k]	[-10k, -1k]	[-1k, -100]	[-100, -10]	[-10, 10]	(10, 100]	(100, 1k]	(1k, 10k]	(10k, 100k]	> 100k	> 0	No Trade
< -100k	885	18.1%	21.3%	9.1%	2.3%	0.4%	0.5%	0.4%	1.4%	8.2%	21.5%	16.8%	48.7%	36.8%
[-100k, -10k]	9,465	2.0%	20.5%	25.6%	8.2%	1.9%	4.7%	1.5%	5.4%	14.0%	14.1%	2.0%	40.0%***	34.3%
[-10k, -1k]	67,616	0.1%	3.7%	22.9%	20.1%	9.0%	12.2%	5.2%	10.8%	13.3%	2.7%	0.1%	39.4%***	41.9%
[-1k, -100]	197,683	0.0%	0.3%	6.4%	25.8%	19.7%	17.8%	11.3%	13.8%	4.5%	0.4%	0.0%	39.5%***	45.3%
[-100, -10]	437,650	0.0%	0.0%	0.9%	7.1%	24.5%	46.9%	15.4%	4.5%	0.7%	0.1%	0.0%	44.2%***	42.7%
[-10, 10]	3,258,652	0.0%	0.0%	0.2%	0.8%	5.1%	89.3%	3.8%	0.6%	0.1%	0.0%	0.0%	57.4%***	36.3%
(10, 100]	307,017	0.0%	0.0%	1.2%	8.3%	23.0%	38.9%	21.7%	6.0%	0.8%	0.1%	0.0%	51.9%***	31.0%
(100, 1k]	132,354	0.0%	0.4%	7.1%	23.4%	12.9%	15.4%	12.4%	21.0%	6.8%	0.5%	0.0%	50.0%	31.2%
(1k, 10k]	49,767	0.1%	3.9%	20.8%	12.9%	4.1%	10.6%	4.0%	12.7%	25.3%	5.5%	0.1%	54.3%***	23.8%
(10k, 100k]	10,482	1.9%	17.0%	14.1%	3.7%	1.6%	3.6%	1.5%	3.8%	19.2%	30.4%	3.1%	60.6%***	20.0%
> 100k	1,063	17.8%	15.5%	6.3%	1.2%	0.1%	0.2%	0.2%	0.3%	5.8%	22.8%	29.7%	58.9%***	16.7%

Panel B: 3-Month Horizon

Starting Group	N	< -100k	[-100k, -10k]	[-10k, -1k]	[-1k, -100]	[-100, -10]	[-10, 10]	(10, 100]	(100, 1k]	(1k, 10k]	(10k, 100k]	> 100k	> 0	No Trade
< -100k	679	14.6%	18.4%	17.3%	3.1%	2.4%	0.7%	0.3%	3.4%	6.5%	21.4%	11.9%	43.9%**	56.7%
[-100k, -10k]	7,164	2.3%	16.3%	22.4%	9.9%	2.8%	4.3%	2.0%	8.0%	17.0%	12.5%	2.5%	44.8%***	52.4%
[-10k, -1k]	45,076	0.2%	3.5%	19.2%	18.4%	8.7%	15.9%	7.0%	11.5%	12.4%	3.1%	0.2%	44.3%***	52.5%
[-1k, -100]	134,578	0.0%	0.6%	6.2%	20.7%	17.6%	24.5%	12.5%	12.4%	4.8%	0.6%	0.0%	44.7%***	58.2%
[-100, -10]	314,316	0.0%	0.1%	1.2%	6.6%	20.0%	53.0%	13.6%	4.4%	1.0%	0.1%	0.0%	47.6%***	57.7%
[-10, 10]	2,766,421	0.0%	0.0%	0.4%	1.0%	4.8%	88.9%	3.8%	0.8%	0.3%	0.0%	0.0%	60.2%***	52.6%
(10, 100]	230,988	0.0%	0.1%	1.2%	6.7%	17.1%	51.0%	17.5%	5.4%	1.1%	0.1%	0.0%	54.1%***	49.3%
(100, 1k]	94,674	0.0%	0.6%	6.5%	17.1%	13.7%	23.4%	12.7%	17.7%	7.4%	0.8%	0.0%	52.9%***	47.7%
(1k, 10k]	36,798	0.2%	3.7%	15.7%	14.0%	6.5%	11.8%	5.4%	13.7%	22.6%	6.1%	0.2%	55.6%***	38.7%
(10k, 100k]	7,568	2.0%	13.5%	15.1%	6.3%	2.0%	4.0%	1.7%	5.5%	19.0%	27.0%	4.0%	60.1%***	33.2%
> 100k	793	11.3%	14.4%	6.1%	4.0%	0.2%	0.2%	0.4%	1.7%	9.8%	29.9%	22.0%	64.0%***	39.7%

Risk-adjusted persistence: a daily-frequency analog

The risk-adjusted transition matrix below buckets each user-month (with ≥ 10 active days) by an annualized ratio built from daily PnL changes *within* that month:

Sharpe analog (daily, annualized)

$$\sqrt{365} \frac{\overline{\Delta \text{PnL}_d}}{\sigma(\Delta \text{PnL}_d)}$$

The Sortino analog replaces the denominator with the downside semi-deviation of ΔPnL_d . Distinct from the monthly user-level ratios earlier: here the frequency is daily and the unit is the user-month.

Weak persistence survives risk adjustment (Sharpe analog)

Panel A: 1-Month Horizon

Starting Group	N	< -10	[-10, -5)	[-5, -3)	[-3, -1)	[-1, 1]	(1, 3]	(3, 5]	(5, 10]	> 10	> 0	No Trade
< -10	51,081	5.7%	19.5%	17.7%	18.3%	20.4%	9.1%	5.6%	3.2%	0.5%	28.0%***	45.6%
[-10, -5)	503,854	1.1%	12.7%	18.6%	19.7%	24.6%	11.3%	7.6%	4.0%	0.5%	34.9%***	44.6%
[-5, -3)	749,270	0.5%	9.3%	18.1%	17.9%	27.4%	12.1%	9.8%	4.4%	0.4%	39.9%***	41.5%
[-3, -1)	754,878	0.4%	7.2%	13.0%	17.6%	35.2%	14.4%	7.8%	4.1%	0.3%	45.0%***	24.8%
[-1, 1]	1,447,993	0.2%	4.0%	7.9%	13.6%	54.9%	11.2%	5.3%	2.9%	0.2%	45.8%***	25.4%
(1, 3]	576,790	0.3%	6.2%	12.7%	18.4%	29.9%	14.6%	10.8%	6.5%	0.6%	45.7%***	25.5%
(3, 5]	436,108	0.3%	6.5%	15.8%	14.2%	22.0%	15.7%	15.3%	9.2%	1.0%	52.4%***	33.4%
(5, 10]	295,852	0.4%	6.5%	12.7%	12.6%	17.1%	16.1%	16.3%	15.3%	2.9%	59.9%***	36.2%
> 10	43,737	0.6%	7.5%	11.5%	10.2%	9.9%	9.3%	10.8%	22.9%	17.2%	65.3%***	33.1%

Panel B: 3-Month Horizon

Starting Group	N	< -10	[-10, -5)	[-5, -3)	[-3, -1)	[-1, 1]	(1, 3]	(3, 5]	(5, 10]	> 10	> 0	No Trade
< -10	35,570	1.8%	10.3%	17.1%	18.6%	31.6%	9.1%	7.1%	3.9%	0.4%	35.1%***	60.3%
[-10, -5)	392,401	0.7%	9.4%	16.8%	17.5%	29.5%	11.8%	8.8%	5.0%	0.5%	39.7%***	62.4%
[-5, -3)	605,011	0.3%	7.9%	16.1%	16.7%	31.1%	12.4%	10.1%	4.9%	0.5%	42.6%***	60.3%
[-3, -1)	624,556	0.3%	6.0%	13.1%	17.3%	38.1%	12.5%	8.2%	4.2%	0.3%	42.9%***	49.8%
[-1, 1]	1,217,754	0.2%	3.8%	8.7%	13.7%	51.6%	12.1%	6.3%	3.3%	0.3%	47.0%***	53.5%
(1, 3]	476,227	0.3%	5.4%	11.5%	14.3%	35.4%	15.4%	10.7%	6.3%	0.7%	51.0%***	49.0%
(3, 5]	351,788	0.3%	6.2%	13.9%	14.2%	25.6%	15.6%	14.1%	8.9%	1.3%	52.9%***	51.9%
(5, 10]	237,151	0.3%	5.9%	11.9%	13.1%	21.4%	16.0%	14.3%	14.3%	2.8%	58.4%***	53.2%
> 10	29,329	0.5%	5.9%	10.9%	10.9%	15.9%	13.7%	13.4%	18.8%	9.9%	63.9%***	49.4%

Decomposing losses: the bid-ask spread

Improve each taker trade's price by the *minimum* effective half-spread, \$0.005 (standard tick) or \$0.0005 (small-tick regime), and adjust the maker's price identically, so aggregate PnL stays zero. This gives a counterfactual PnL *excluding* (a lower bound on) spread costs.

Fraction of a bucket's gain/loss attributable to the spread

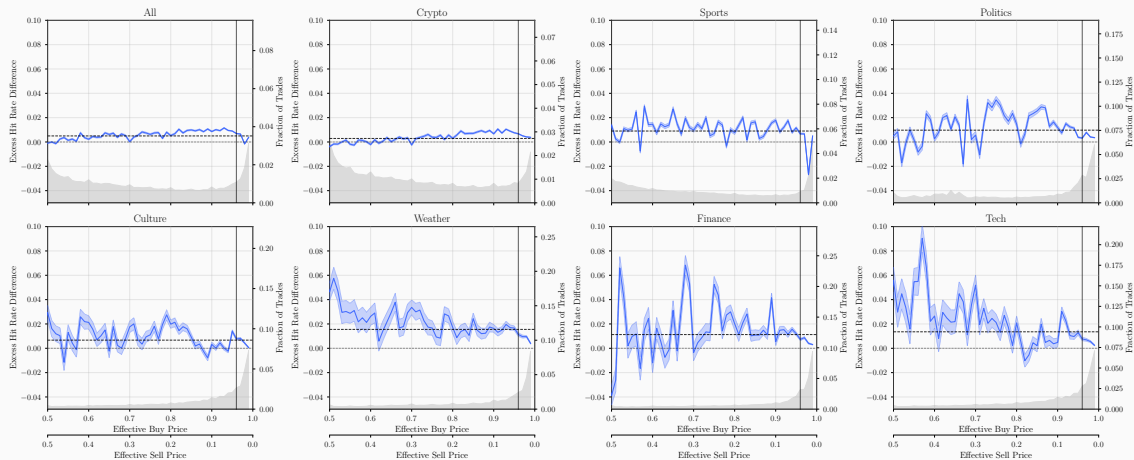
$$\text{Frac. Spread} = \frac{\overline{\text{PnL}}_{\text{excl.}} - \overline{\text{PnL}}_{\text{incl.}}}{|\overline{\text{PnL}}_{\text{incl.}}|}$$

Because we remove only the tick floor, this is a lower bound on the role of the spread.

Spread costs explain a meaningful share of losses

Percentile	N Users	Mean Trades	Mean PnL (incl.)	Mean PnL (excl.)	Difference	Frac. Spread	Maker %
Panel A: Losers (PnL < 0)							
0-20	340,794	30	-0.38	0.34	-0.72***	190.9%	12.9%
20-40	340,794	53	-3.25	0.38	-3.63***	111.7%	11.0%
40-60	340,794	85	-13.62	-5.51	-8.11***	59.6%	10.1%
60-80	340,794	160	-61.01	-50.78	-10.23***	16.8%	14.0%
80-90	170,397	437	-256.94	-236.27	-20.67***	8.0%	17.9%
90-95	85,198	788	-893.47	-860.78	-32.70***	3.7%	28.6%
95-99	68,159	1,468	-2,762.57	-2,692.55	-70.02***	2.5%	35.4%
99-99.9	15,336	4,444	-16,682.47	-16,360.21	-322.26***	1.9%	29.2%
99.9-100	1,703	13,753	-269,192.06	-266,781.77	-2,410.29	0.9%	37.2%
Panel B: Winners (PnL > 0)							
0-20	152,885	29	0.19	0.49	-0.31***	163.2%	24.1%
20-40	152,885	54	2.17	3.15	-0.97***	44.9%	23.2%
40-60	152,885	97	13.36	15.24	-1.88***	14.1%	20.5%
60-80	152,885	191	72.58	75.30	-2.72***	3.8%	24.9%
80-90	76,442	600	345.54	353.68	-8.14***	2.4%	26.3%
90-95	38,221	1,292	1,308.31	1,304.11	4.21	-0.3%	35.1%
95-99	30,577	4,145	5,111.42	5,047.34	64.07***	-1.3%	40.6%
99-99.9	6,880	37,322	38,633.66	37,179.87	1,453.79***	-3.8%	44.9%
99.9-100	764	255,975	703,059.84	680,242.46	22,817.38***	-3.2%	49.3%

Makers out-predict takers, concentrated at extreme prices



A Simulation Benchmark

Simulation benchmark: markets and traders

A parsimonious repeated binary-market economy. Prices are unbiased in log-odds with small residual mispricings; $\Lambda(x) = (1 + e^{-x})^{-1}$.

M markets

$$\theta_m \sim \text{U}(0, 1), \quad Y_m \sim \text{Bern}(\theta_m)$$

$$z_m = \log(\theta_m / (1 - \theta_m))$$

$$\ell_m = z_m + u_m, \quad u_m \sim N(0, \sigma_p^2)$$

$$p_m = \Lambda(\ell_m)$$

N traders: 4 independent primitives

- wealth $\log W_{i0} = \alpha_W + \varepsilon_i^W$
- true precision $\log \tau_i = \alpha_\tau + \varepsilon_i^\tau$
- maker propensity $\mu_i = \Lambda(\alpha_\mu + \varepsilon_i^\mu)$
- overconfidence $\log \hat{\tau}_i = \log \tau_i + b_i$

Independent draws: the high-wealth, high-skill, high-maker, and overconfident tails are different traders.

Simulation benchmark: beliefs, demand, execution

Signal and posterior belief

$$x_{im} = z_m + \eta_{im}, \quad \eta_{im} \sim N(0, 1/\tau_i)$$

$$\tilde{z}_{im} = \lambda_i x_{im} + (1 - \lambda_i) \ell_m$$

$$\lambda_i = \frac{\hat{\tau}_i}{\hat{\tau}_i + \tau_0}, \quad q_{im} = \Lambda(\tilde{z}_{im})$$

Demand (Kelly-like, wealth-capped)

$$d_{im}^* = \kappa_i W_{it} \frac{q_{im} - p_m}{p_m(1 - p_m)}, \quad \kappa_i = \kappa_0 \hat{\tau}_i^\gamma$$

Maker-taker execution

- maker w.p. μ_i ; constant half-spread h_0
- ask $p_m + h_0$, bid $p_m - h_0$
- makers earn the spread; per-market PnL $\approx$ zero-sum

PnL and survival

- $\pi_{im} = Y_m - p_{im}^{\text{exec}}$ (reference buyers)
- $W_{i,t+1} = W_{it} + \Pi_{im}$; exit if $W_{it} \leq W_{\min}$
- wealth-dependent participation: winners compound, losers exit

Three channels jointly reproduce the empirical patterns

Parsimonious repeated binary-market economy; traders differ in wealth, true and perceived signal precision, and maker propensity. A structured “null”, not a calibration.

Scenario	Fraction Losers	Top 1% Gains	Top 5% Gains	Maker Sh. Top 1%	Bottom 1% Losses	Bottom 5% Losses	EHR Top–Bot (5% Groups)	Losers Flipped (No Spread)
Empirical Facts	69.0%	76.5%	91.4%	42.0%	68.1%	86.1%	0.280	18.5%
Simulation Baseline	52.0%	30.2%	58.2%	34.1%	32.3%	59.0%	0.006	0.5%
+ Skilled Traders	67.4%	42.7%	70.3%	46.5%	35.4%	61.8%	0.027	0.5%
+ Maker Specialization	61.6%	45.4%	70.6%	70.0%	32.5%	59.7%	0.009	5.3%
+ Overconfidence	55.7%	37.9%	65.0%	33.8%	31.7%	58.9%	0.006	0.4%
+ All Three	78.1%	76.0%	89.4%	66.0%	28.3%	55.1%	0.029	3.4%

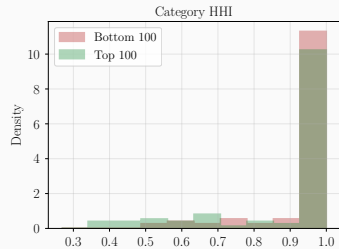
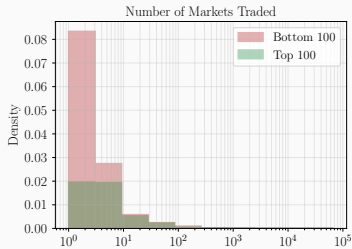
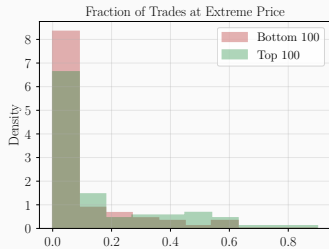
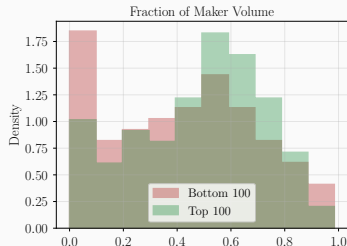
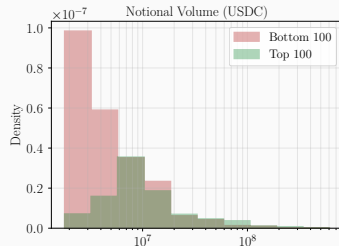
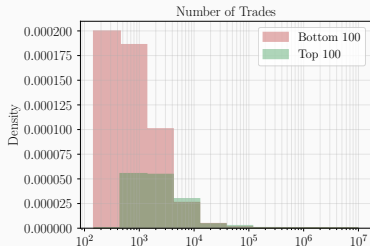
Trading Behavior of the Most Profitable Users

Top 12 winners and losers (of the top 100)

Winners							Losers						
Rank	PnL (\$M)	Trades	Volume (\$M)	Maker vol. %	Extreme price %	Top category	Rank	PnL (\$M)	Trades	Volume (\$M)	Maker vol. %	Extreme price %	Top category
1	22.0	19,695	43.0	45	0	Politics	1	-10.0	19,531	150.3	31	0	Sports
2	16.6	30,201	76.6	39	15	Politics	2	-7.2	51,466	434.8	1	0	Sports
3	10.4	162,330	265.5	73	4	Sports	3	-6.8	1,254	19.1	70	60	Politics
4	8.7	6,649	16.4	40	0	Politics	4	-5.5	62,537	122.9	63	5	Sports
5	7.7	18,074	40.6	29	0	Politics	5	-5.0	2,708	13.4	36	0	Sports
6	7.5	7,646	14.0	48	0	Politics	6	-5.0	1,493	13.5	80	9	Politics
7	6.5	1,749,437	332.4	82	13	Sports	7	-4.8	120,669	62.1	92	35	Politics
8	6.1	8,592	23.5	46	0	Politics	8	-4.7	6,503	20.1	57	22	Politics
9	5.9	5,793	32.2	75	6	Politics	9	-4.5	10,987	37.2	29	2	Sports
10	5.7	4,614	20.1	29	0	Sports	10	-3.8	10,093	49.9	34	2	Sports
11	5.6	7,563	11.2	51	0	Politics	11	-3.7	7,815	10.2	87	32	Politics
12	5.6	3,263,133	604.7	18	9	Sports	12	-3.7	29,230	144.3	32	2	Sports

Gains span many sports teams (81%) and 2024 US election markets (15%); many top earners profit mainly from [liquidity provision](#).

Winners and losers look different in behavior, not just luck



Are the top traders insiders?

- Top-100 gains span [many distinct sports leagues](#) and a handful of [2024 US election](#) markets.
- Within a trader, profits are usually concentrated in [one or a few markets](#); a meaningful fraction earn the bulk from [liquidity provision](#), not directional bets.
- For insider information to explain these profits, one trader would need private knowledge across many sports/teams, or knowledge of millions of votes before they were cast.
- [We do not claim there is no insider trading](#), but it does not look like the dominant source of the largest winners' profits.

Conclusion

Conclusion

1. Profits are highly concentrated; most users lose money (near-zero-sum).
2. Prices are relatively efficient and informative on aggregate.
3. Per-trade accuracy declines through the PnL distribution, but monthly performance is only weakly persistent.
4. Liquidity provision is the strongest predictor of avoiding loss.
5. The most successful users' gains look like skill / liquidity provision, not insider trading.

Takeaway

Price efficiency and extreme distributional inequality coexist. Treat prediction markets as competitive financial markets: fair prices, zero-sum PnL, higher loss probability for liquidity takers.

Thank you!



Data



Paper

References

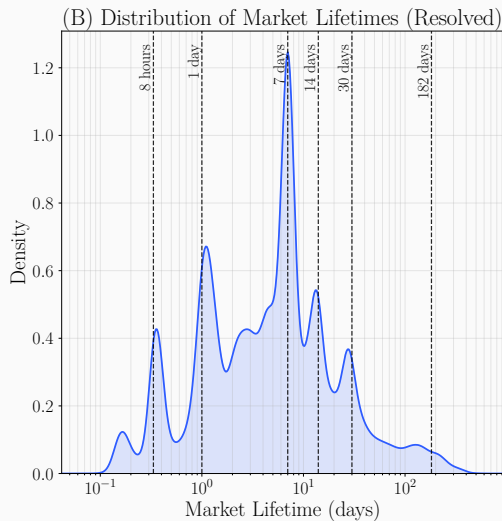
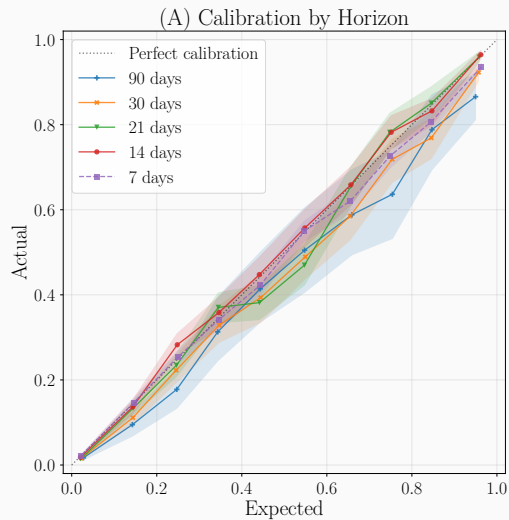
References i

- Brad M. Barber and Terrance Odean. Trading is hazardous to your wealth: The common stock investment performance of individual investors. *Journal of Finance*, 55(2):773–806, 2000. doi: 10.1111/0022-1082.00226.
- Brad M. Barber, Yi-Tsung Lee, Yu-Jane Liu, and Terrance Odean. Just how much do individual investors lose by trading? *Review of Financial Studies*, 22(2):609–632, 2009. doi: 10.1093/rfs/hhn046.
- Jonah Becker. The microstructure of wealth transfer in prediction markets.
<https://www.jbecker.dev/research/prediction-market-microstructure>, 2025. Accessed: 2026-03-15.
- Michaël Dewally, Louis H. Ederington, and Chitru S. Fernando. Determinants of trader profits in commodity futures markets. *Review of Financial Studies*, 26(10):2648–2683, 2013. doi: 10.1093/rfs/hht048.
- Robert Forsythe, Forrest Nelson, George R. Neumann, and Jack Wright. Anatomy of an experimental political stock market. *American Economic Review*, 82(5):1142–1161, 1992.
- Roberto Gomez Cram, Yunhan Guo, Theis Ingerslev Jensen, and Howard Kung. Financial prediction markets: A new measure of earnings expectations. *Working paper*, 2025. doi: 10.2139/ssrn.5933475.
- Roberto Gomez Cram, Yunhan Guo, Theis Ingerslev Jensen, and Howard Kung. Prediction market accuracy: Crowd wisdom or informed minority? *Working paper*, 2026. doi: 10.2139/ssrn.6617059.

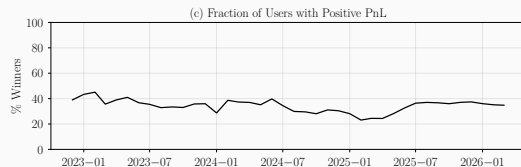
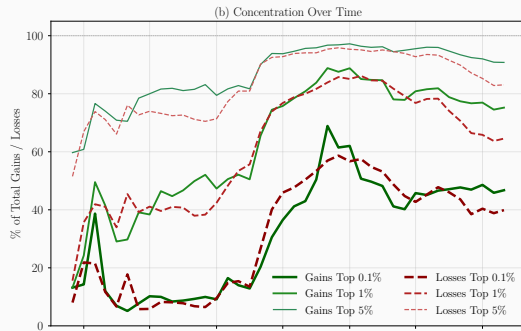
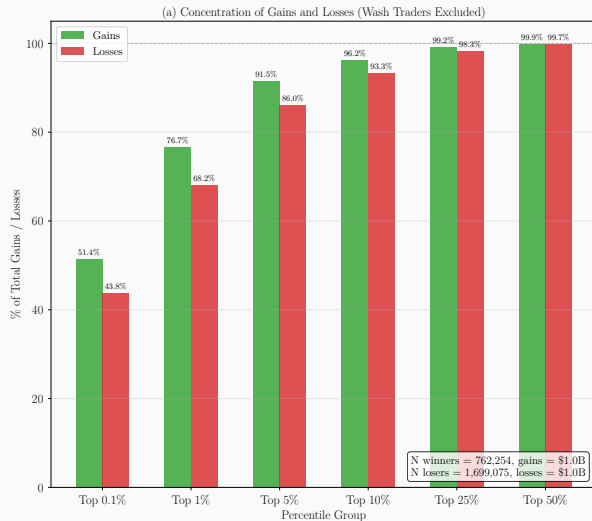
- Friedrich A. Hayek. The use of knowledge in society. *American Economic Review*, 35(4):519–530, 1945.
- Jianfeng Hu, Antonia Kirilova, Seongkyu (Gilbert) Park, and Doojin Ryu. Who profits from trading options? *Management Science*, 70(7):4742–4761, 2024. doi: 10.1287/mnsc.2023.4916.
- Charles F. Manski. Interpreting the predictions of prediction markets. *Economics Letters*, 91(3):425–429, 2006. doi: 10.1016/j.econlet.2006.01.004.
- Hunter Ng, Lin Peng, Yubo Tao, and Dexin Zhou. Price discovery and trading in modern prediction markets. *Working paper*, 2026. doi: 10.2139/ssrn.5331995.
- Terrance Odean. Do investors trade too much? *American Economic Review*, 89(5):1279–1298, 1999.
- Kenneth Oliven and Thomas A. Rietz. Suckers are born but markets are made: Individual rationality, arbitrage, and market efficiency on an electronic futures market. *Management Science*, 50(3):336–351, 2004. doi: 10.1287/mnsc.1040.0191.
- Allen Sirolli, Hongyao Ma, Yash Kanoria, and Rajiv Sethi. Network-based detection of wash trading. *Working paper*, 2025. doi: 10.2139/ssrn.5714122.

- Erik Snowberg, Justin Wolfers, and Eric Zitzewitz. Prediction markets for economic forecasting. In Graham Elliott and Allan Timmermann, editors, *Handbook of Economic Forecasting*, volume 2, pages 657–687. Elsevier, 2013. doi: 10.1016/B978-0-444-53683-9.00011-6.
- Kwok Ping Tsang and Zichao Yang. The anatomy of Polymarket: Evidence from the 2024 presidential election. *Working paper*, 2026. doi: 10.48550/arXiv.2603.03136.
- Justin Wolfers and Eric Zitzewitz. Prediction markets. *Journal of Economic Perspectives*, 18(2):107–126, 2004. doi: 10.1257/0895330041371321.

Appendix: Calibration at longer horizons



Appendix: Concentration excluding wash trading



Appendix: Probit excluding likely wash traders

	All Users				Users With More Than 100 Trades		Users With More Than 1,000 Trades	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Frac Extreme Price	-0.012*** (0.001)	-0.017*** (0.001)	-0.007*** (0.001)	-0.003*** (0.001)	0.002 (0.002)	0.007*** (0.002)	-0.129*** (0.006)	-0.128*** (0.006)
Frac Maker Volume		-0.354*** (0.001)	-0.349*** (0.001)	-0.343*** (0.001)	-0.318*** (0.003)	-0.314*** (0.003)	-0.142*** (0.006)	-0.137*** (0.006)
Log N Trades		-0.001*** (0.000)	0.004*** (0.000)	0.018*** (0.000)	0.030*** (0.001)	0.034*** (0.001)	-0.002 (0.002)	-0.005** (0.002)
Log Total Volume		0.020*** (0.000)	0.018*** (0.000)	0.020*** (0.000)	-0.022*** (0.000)	-0.023*** (0.001)	-0.044*** (0.001)	-0.041*** (0.001)
Category HHI			0.080*** (0.001)	-0.057*** (0.002)	-0.019*** (0.003)	-0.080*** (0.004)	-0.136*** (0.007)	-0.146*** (0.010)
Counterparty HHI			-0.002 (0.002)	0.040*** (0.002)	0.354*** (0.010)	0.403*** (0.011)	0.827*** (0.063)	0.812*** (0.063)
Traded Crypto				-0.074*** (0.001)		-0.018*** (0.002)		0.056*** (0.006)
Traded Finance				-0.034*** (0.001)		-0.019*** (0.002)		-0.009* (0.005)
Traded Politics				-0.056*** (0.001)		-0.000 (0.002)		0.041*** (0.005)
Traded Tech				-0.034*** (0.001)		-0.023*** (0.002)		-0.026*** (0.005)
Traded Culture				0.002** (0.001)		-0.005** (0.002)		-0.023*** (0.005)
Traded Weather				-0.036*** (0.001)		-0.024*** (0.002)		-0.008* (0.005)
Observations	2,472,863	2,472,863	2,472,863	2,472,863	470,965	470,965	78,223	78,223
Pseudo R ²	0.000	0.034	0.036	0.043	0.042	0.043	0.055	0.057

Appendix: Transition matrix, 6- and 12-month horizons

Panel A: 6-Month Horizon

Starting Group	N	< -100k	[-100k, -10k]	[-10k, -1k]	[-1k, -100]	[-100, -10]	[-10, 10]	(10, 100]	(100, 1k]	(1k, 10k]	(10k, 100k]	> 100k	> 0	No Trade
< -100k	492	11.6%	21.9%	9.0%	7.7%	0.6%	1.3%	1.9%	3.9%	13.5%	21.9%	6.5%	49.0%	68.5%
[-100k, -10k]	4,889	2.0%	14.1%	21.4%	10.9%	2.9%	5.4%	2.8%	8.3%	16.1%	13.8%	2.3%	46.7%***	61.6%
[-10k, -1k]	25,923	0.3%	3.8%	19.7%	18.3%	7.5%	12.4%	5.1%	13.3%	14.8%	4.4%	0.3%	44.0%***	62.8%
[-1k, -100]	78,846	0.0%	0.8%	6.9%	20.0%	16.8%	23.9%	12.0%	12.8%	5.8%	0.9%	0.0%	44.9%***	67.4%
[-100, -10]	206,864	0.0%	0.1%	1.5%	6.9%	17.2%	56.3%	11.7%	4.9%	1.2%	0.1%	0.0%	51.2%***	69.2%
[-10, 10]	2,095,300	0.0%	0.0%	1.2%	1.8%	5.2%	85.9%	4.1%	1.1%	0.4%	0.1%	0.0%	60.7%***	63.5%
(10, 100]	153,867	0.0%	0.2%	1.8%	6.4%	14.7%	54.4%	14.9%	5.9%	1.6%	0.2%	0.0%	56.6%***	64.1%
(100, 1k]	56,920	0.0%	0.7%	6.8%	16.2%	13.7%	23.3%	12.1%	17.1%	8.8%	1.2%	0.0%	53.5%***	58.7%
(1k, 10k]	21,496	0.2%	3.8%	16.2%	14.5%	5.5%	9.4%	4.9%	14.2%	23.4%	7.5%	0.4%	54.8%***	47.8%
(10k, 100k]	4,878	2.0%	11.6%	14.2%	7.8%	1.9%	4.5%	1.6%	6.2%	19.0%	26.9%	4.3%	61.2%***	42.3%
> 100k	526	5.3%	14.5%	8.8%	3.1%	0.9%	0.9%	0.4%	1.3%	15.9%	25.1%	23.8%	67.0%***	56.8%

Panel B: 12-Month Horizon

Starting Group	N	< -100k	[-100k, -10k]	[-10k, -1k]	[-1k, -100]	[-100, -10]	[-10, 10]	(10, 100]	(100, 1k]	(1k, 10k]	(10k, 100k]	> 100k	> 0	No Trade
< -100k	293	7.0%	18.3%	11.3%	1.4%	2.8%	5.6%	2.8%	5.6%	16.9%	26.8%	1.4%	56.3%	75.8%
[-100k, -10k]	2,796	1.4%	14.4%	18.6%	10.7%	3.7%	3.8%	3.9%	10.5%	17.3%	13.5%	2.3%	50.0%	71.7%
[-10k, -1k]	13,713	0.5%	4.7%	17.1%	19.9%	10.0%	9.6%	6.1%	13.3%	13.3%	5.1%	0.5%	43.1%***	71.4%
[-1k, -100]	42,077	0.1%	0.8%	7.2%	19.0%	16.6%	24.8%	11.9%	12.0%	6.2%	1.4%	0.1%	46.1%***	73.3%
[-100, -10]	122,436	0.0%	0.2%	3.1%	8.8%	14.8%	57.4%	9.6%	4.3%	1.5%	0.3%	0.0%	46.2%***	68.1%
[-10, 10]	1,023,638	0.0%	0.1%	1.2%	3.3%	7.0%	81.5%	4.5%	1.9%	0.5%	0.1%	0.0%	58.0%***	69.3%
(10, 100]	91,688	0.0%	0.1%	3.3%	8.2%	13.7%	55.0%	11.8%	6.0%	1.5%	0.4%	0.0%	50.4%	66.0%
(100, 1k]	32,153	0.0%	0.9%	7.1%	15.5%	13.8%	23.0%	12.7%	15.9%	9.1%	1.8%	0.1%	54.3%***	67.5%
(1k, 10k]	11,109	0.3%	4.1%	14.4%	16.4%	6.8%	8.1%	5.4%	14.3%	20.4%	9.1%	0.8%	53.7%***	59.2%
(10k, 100k]	2,604	1.6%	11.4%	15.3%	8.8%	1.7%	2.1%	1.4%	8.5%	19.4%	24.7%	5.1%	60.5%***	54.2%
> 100k	302	8.0%	14.3%	8.9%	5.4%	0.9%	3.6%	1.8%	6.2%	12.5%	25.0%	13.4%	61.6%**	62.9%

Appendix: Transition matrix, makers only

Panel A: 1-Month Horizon

Starting Group	N	< -100k	[-100k, -10k)	[-10k, -1k)	[-1k, -100)	[-100, -10)	[-10, 10]	(10, 100]	(100, 1k)	(1k, 10k]	(10k, 100k]	> 100k	> 0	No Trade
< -100k	503	19.4%	27.5%	5.6%	2.5%	0.0%	0.3%	0.9%	1.5%	6.2%	21.3%	14.8%	44.8%*	35.6%
[-100k, -10k)	5,154	2.4%	20.6%	23.9%	6.3%	1.5%	1.4%	1.3%	5.9%	16.5%	17.6%	2.6%	44.4%***	31.6%
[-10k, -1k)	34,990	0.1%	4.3%	23.1%	18.4%	7.3%	7.4%	5.9%	13.1%	16.6%	3.7%	0.1%	44.0%***	46.9%
[-1k, -100)	81,780	0.0%	0.5%	8.0%	24.7%	15.4%	13.5%	12.7%	17.6%	6.9%	0.6%	0.0%	45.1%***	49.4%
[-100, -10)	149,589	0.0%	0.1%	1.4%	9.0%	23.5%	41.6%	15.6%	7.3%	1.5%	0.1%	0.0%	43.1%***	54.3%
[-10, 10]	586,064	0.0%	0.0%	0.5%	2.3%	9.5%	77.0%	8.6%	1.6%	0.5%	0.0%	0.0%	57.5%***	54.5%
(10, 100]	141,020	0.0%	0.1%	1.6%	9.8%	19.6%	34.0%	24.4%	8.7%	1.7%	0.1%	0.0%	54.7%***	47.6%
(100, 1k]	75,114	0.0%	0.5%	6.9%	19.8%	12.0%	13.5%	13.4%	24.1%	9.1%	0.7%	0.0%	55.1%***	41.5%
(1k, 10k]	36,297	0.1%	3.8%	16.9%	12.9%	4.7%	6.8%	4.3%	15.2%	29.1%	6.0%	0.2%	59.6%***	39.5%
(10k, 100k]	6,501	2.3%	16.0%	13.9%	3.9%	1.4%	1.3%	1.1%	4.6%	22.5%	30.0%	3.1%	61.6%***	24.8%
> 100k	586	17.7%	16.7%	4.6%	0.4%	0.4%	0.2%	0.0%	1.3%	6.5%	29.7%	22.4%	59.9%***	19.1%

Panel B: 3-Month Horizon

Starting Group	N	< -100k	[-100k, -10k)	[-10k, -1k)	[-1k, -100)	[-100, -10)	[-10, 10]	(10, 100]	(100, 1k)	(1k, 10k]	(10k, 100k]	> 100k	> 0	No Trade
< -100k	392	10.3%	20.0%	16.4%	5.5%	0.0%	0.6%	0.6%	3.6%	7.9%	26.7%	8.5%	47.3%	57.9%
[-100k, -10k)	3,925	2.6%	15.7%	22.8%	8.2%	2.2%	1.7%	2.3%	7.0%	19.3%	15.6%	2.6%	47.9%*	50.0%
[-10k, -1k)	23,344	0.3%	4.3%	18.5%	18.4%	7.7%	9.8%	7.3%	13.4%	15.7%	4.4%	0.3%	47.5%***	55.1%
[-1k, -100)	54,611	0.0%	0.9%	7.9%	20.4%	14.1%	16.0%	14.7%	16.8%	7.9%	1.2%	0.1%	49.9%	63.7%
[-100, -10)	110,055	0.0%	0.1%	2.0%	8.9%	18.5%	43.6%	16.2%	8.2%	2.1%	0.2%	0.0%	51.0%***	74.1%
[-10, 10]	472,305	0.0%	0.0%	0.5%	1.9%	9.4%	77.1%	8.7%	1.7%	0.5%	0.0%	0.0%	61.4%***	72.9%
(10, 100]	101,713	0.0%	0.1%	2.0%	8.5%	14.8%	43.9%	19.9%	8.4%	2.1%	0.2%	0.0%	56.0%***	66.6%
(100, 1k]	52,780	0.0%	0.9%	7.5%	17.1%	12.5%	16.5%	13.2%	20.9%	10.0%	1.3%	0.1%	55.4%***	58.3%
(1k, 10k]	26,614	0.2%	4.0%	15.2%	14.0%	6.5%	9.0%	6.0%	14.8%	23.0%	7.0%	0.3%	57.3%***	51.3%
(10k, 100k]	4,718	1.8%	14.6%	15.5%	6.0%	1.8%	0.8%	1.4%	6.4%	21.3%	26.7%	3.7%	59.8%***	37.9%
> 100k	434	15.6%	16.4%	8.2%	2.0%	0.4%	0.8%	0.0%	3.5%	9.0%	25.8%	18.4%	56.6%**	41.0%

Appendix: Risk-adjusted transition (Sortino / downside-vol)

Panel A: 1-Month Horizon

Starting Group	N	< -10	[-10, -5)	[-5, -3)	[-3, -1)	[-1, 1]	(1, 3]	(3, 5]	(5, 10]	> 10	> 0	No Trade
< -10	34,140	6.0%	27.0%	18.6%	15.9%	13.7%	6.5%	2.9%	3.6%	5.8%	25.2%***	45.8%
[-10, -5)	602,292	0.7%	17.1%	21.5%	18.8%	18.4%	8.2%	4.0%	5.0%	6.4%	31.9%***	46.0%
[-5, -3)	823,025	0.3%	12.2%	21.2%	18.2%	21.9%	9.5%	4.5%	5.4%	6.9%	36.5%***	44.8%
[-3, -1)	709,991	0.2%	8.5%	15.0%	16.8%	29.4%	13.5%	4.9%	5.5%	6.3%	45.5%***	28.8%
[-1, 1]	1,009,087	0.1%	5.1%	9.5%	16.5%	46.9%	11.0%	3.4%	3.5%	4.0%	44.0%***	28.8%
(1, 3]	396,884	0.1%	7.8%	13.1%	21.6%	29.6%	9.3%	5.1%	6.4%	6.9%	39.3%***	27.5%
(3, 5]	192,510	0.2%	8.8%	16.3%	18.3%	18.9%	10.8%	6.9%	9.0%	10.9%	46.3%***	27.6%
(5, 10]	248,619	0.2%	9.4%	15.9%	15.2%	17.7%	11.1%	7.0%	10.1%	13.3%	50.2%*	29.4%
> 10	351,165	0.3%	9.7%	17.0%	14.2%	15.3%	10.0%	6.6%	10.0%	16.9%	50.8%***	33.5%

Panel B: 3-Month Horizon

Starting Group	N	< -10	[-10, -5)	[-5, -3)	[-3, -1)	[-1, 1]	(1, 3]	(3, 5]	(5, 10]	> 10	> 0	No Trade
< -10	23,420	1.6%	13.2%	20.3%	19.0%	26.1%	7.3%	3.1%	3.8%	5.7%	31.5%***	61.1%
[-10, -5)	469,837	0.5%	13.5%	19.8%	17.9%	22.4%	8.9%	4.5%	5.6%	7.0%	35.7%***	64.2%
[-5, -3)	667,941	0.2%	10.3%	19.9%	17.8%	24.6%	9.8%	4.7%	5.3%	7.3%	38.5%***	63.5%
[-3, -1)	588,798	0.2%	7.9%	15.8%	18.3%	31.7%	10.6%	4.3%	4.9%	6.3%	40.3%***	54.7%
[-1, 1]	851,932	0.1%	5.1%	10.7%	16.6%	43.1%	11.9%	3.9%	3.9%	4.8%	44.9%***	57.8%
(1, 3]	329,482	0.1%	6.8%	12.7%	16.0%	33.3%	11.9%	5.4%	6.4%	7.3%	47.4%***	53.5%
(3, 5]	158,098	0.2%	7.9%	14.9%	15.6%	24.3%	12.2%	6.6%	8.7%	9.6%	48.6%***	50.5%
(5, 10]	201,133	0.2%	8.0%	15.8%	15.5%	20.6%	11.7%	7.2%	9.6%	11.3%	49.9%	49.7%
> 10	273,344	0.2%	8.6%	16.0%	14.8%	18.8%	10.5%	6.6%	9.6%	14.9%	50.6%***	52.4%

Appendix: Spread decomposition, Politics

Percentile	N Users	Mean Trades	Mean PnL (incl.)	Mean PnL (excl.)	Difference	Frac. Spread	Maker %
Panel A: Losers (PnL < 0)							
0-20	195,444	145	-0.23	0.15	-0.38***	168.3%	20.8%
20-40	195,443	167	-2.13	0.17	-2.30***	108.0%	11.9%
40-60	195,444	225	-9.41	0.33	-9.74***	103.5%	11.8%
60-80	195,443	306	-53.05	-43.82	-9.23***	17.4%	13.1%
80-90	97,722	448	-276.96	-265.02	-11.93***	4.3%	12.2%
90-95	48,861	633	-1,034.74	-1,017.62	-17.11***	1.7%	13.1%
95-99	39,088	1,222	-4,904.59	-4,853.66	-50.94***	1.0%	13.5%
99-99.9	8,795	3,491	-37,200.18	-37,119.84	-80.35	0.2%	15.0%
99.9-100	977	13,378	-442,567.49	-443,957.10	1,389.60**	-0.3%	23.6%
Panel B: Winners (PnL > 0)							
0-20	110,781	104	0.08	0.16	-0.09***	113.0%	26.5%
20-40	110,780	137	0.91	1.26	-0.35***	39.0%	22.1%
40-60	110,780	438	5.04	4.99	0.05	-1.0%	22.6%
60-80	110,780	247	27.29	25.65	1.64***	-6.0%	24.1%
80-90	55,390	458	121.90	119.13	2.78***	-2.3%	21.2%
90-95	27,695	871	448.55	442.60	5.95***	-1.3%	22.4%
95-99	22,156	2,415	2,469.40	2,422.92	46.48***	-1.9%	27.1%
99-99.9	4,986	10,279	23,892.27	23,528.63	363.63***	-1.5%	36.3%
99.9-100	553	28,462	632,805.19	625,071.83	7,733.36***	-1.2%	46.0%

Appendix: Simulation excess-hit-rate grid

